



Revolutionizing Chronic Obstructive Pulmonary Disease Care with the Open AI Application: ChatGPT

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Abstract

Chronic Obstructive Pulmonary Disease (COPD) is one of the major and leading threats to human being. To cope with this health challenge, several studies have been undertaken in the literature. However, COPD is not paid close attention to eliminate it entirely. The current study aims to examine the role of ChatGPT application to bring improvement in controlling and managing COPD in patients. ChatGPT is used to give prompt answers of text-based questions in a variety of fields. It has potential role in knowing the symptoms of COPD, and letting individuals to modify their life styles. ChatGPT suggests medications for COPD individuals based on the established medical guidelines. Compared to the literature, ChatGPT provides a comprehensive list of COPD test and evaluation methods. ChatGPT has the potential to help physicians to take decisions in diagnosing, treating, and managing COPD among individuals in its upcoming versions. More researches can be conducted to identify the limits of ChatGPT application in future works.

Keywords Medication · COPD tests · Pulmonary rehabilitation

Introduction

Proposing and implementing a project on diagnosing and managing Chronic Obstructive Pulmonary Disease (COPD) among patients can bring changes in the lives of people [1]. Since several barriers create challenges for patients and their families to receive an appropriate COPD treatment [2], thus the proposed research can overcome the challenges by helping them to get timely treatment. According to World Health Organization (WHO), COPD is the third leading cause of death worldwide [3]. Globally, high-income countries have shown a decline in the mortalities from COPD. Yet, very little is known about COPD in Latin American, Europe, and Oceania [4]. Organizations such as WHO and United Nations (UN) have not seriously taken steps to address the

COPD in middle- or low-income countries [5]. This is a serious alarm to the world powers and they should be thoughtful to address this health problem.

ChatGPT as a Generative Pre-trained transformed (GPT) is a type of language model that has been recently released by Open AI. It has been trained on a huge amount of dataset and works like being a human in a dialog with users. ChatGPT has been tested and passed by United States Medical Licensing Exam (USMLE) in three exams [6]. The use of ChatGPT is increasing in dealing with health issues [7, 8]. Based on these premises, we used ChatGPT to perform a brief examination of COPD in a scenario of supporting physicians and individuals with the disease.

This research aims to assess the role of ChatGPT in supporting individuals with COPD worldwide. Secure and trustworthy artificial intelligence (AI) applications can serve to regulate and ensure ethics and trustworthiness in the technology for addressing high-risk health issues. Open AI application, ChatGPT is expected to have a greater role in helping clinicians to diagnose and manage COPD among children and old age people and it would be very exciting to observe the benefits that arise from using the ChatGPT for doctors and COPD patients in everyday practice and their lives, respectively.

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Pros and Cons of ChatGPT Application

As a language model, ChatGPT gives us invaluable information about COPD and sources for caregivers and individuals with COPD. ChatGPT is asked to reveal its strengths, while controlling COPD in individuals. In this regard, ChatGPT emphasizes on four areas as follows:

The first area of improvement is to educate the individuals with COPD. ChatGPT can help individuals know symptoms, causes, and various treatment options. This may help the individuals with COPD and physicians understand the disease and make timely decisions to prevent or reduce the impacts of disease. The second area is to bring modifications in lifestyle of individuals having disease. The AI application lets individuals to know the lifestyle changes and manage the COPD symptoms including, daily exercises, giving up smoking, and avoiding triggers like pollution and respiratory infections. ChatGPT suggests individuals to choose the best medication to treat disease and avoid potential side effects. ChatGPT can provide emotional support to individuals and their caregivers by connecting with them and helping them to identify resources and support groups who can cope with the challenges of living with COPD. However, individuals with COPD cannot totally rely on the ChatGPT and they must seek advice from their physicians who could help them by giving proper prescriptions for COPD treatment and also they can provide the personalized recommendations.

One of the limits of ChatGPT is that it gives generic information about disease and resources. However, this might not be a limitation because each person's experience with the disease varies, and treatment plan is tailored to individuals' basic requirements and medical history as well. ChatGPT can still be considered a useful AI application in providing general information about COPD and better understanding the conditions. Despite these advantages, ChatGPT repeats some basic concepts about COPD while asking different text supplied questions about the disease. We asked ChatGPT about treatment plan of COPD. AI application highlights some important steps of a treatment plan to prevent or reduce the consequences of COPD.

ChatGPT suggests a few medicines to treat the disease. Medications include bronchodilators (that relax muscles in the airways), corticosteroids (which aim to reduce inflammation in air ways), and lastly phosphodiesterase-4 inhibitors (that improve lungs functioning by reducing inflammation). Different instruments such as nebulizers, inhalers,

or oral tablets can be used to administer medicines. As answered by ChatGPT, pulmonary rehabilitation is another technique aimed at improving individuals' breathing, resulting in an increase in their endurance and decrease in symptoms. Oxygen therapy is another treatment plan for individuals with COPD, where supplementary oxygen is provided to maintain the oxygen level. Finally, ChatGPT suggests individuals to be vaccinated against the respiratory infections like pneumonia and flue that can help in preventing the exacerbations of the disease.

As a language model, ChatGPT is unable to examine the COPD patients and directly diagnose the individuals with the disease. Only pulmonologists or qualified doctors can diagnose the COPD through a number of tests and evaluations. However, ChatGPT can answer the questions about COPD while treating patients and performing research. It is also rich in the established medical guidelines. ChatGPT can analyze and interpret data received from pulmonary functional tests, images data, blood tests, and patient-reported summaries. More tests and evaluation information can be reported in the upcoming versions of ChatGPT application. To further explore the reported test and evaluations, we search for related articles in PubMed digital library and found that these tests were discussed in various studies. Recent work [9] reports imaging data and pulmonary functional test and their outcomes, but missed the discussion on blood test and patients reports.

ChatGPT has the potential to contribute and create a high impact on bringing improvement in monitoring and controlling COPD and its consequences. Aged people and particularly pregnant and milk-feeding women will have greater and more useful impacts from this application. The Open AI application can help identify and diagnose conditions of COPD and better analyze different types of data to design personalized treatment plans more accurately. ChatGPT can benefit paramedics staff in their decision-making regarding COPD patients by saving their time, energy, and budget that could be spent on testing and other medical tasks.

Chronic Obstructive Pulmonary Disease (COPD) Test and Evaluation Methods

We performed a comparison of tests and evaluation methods reported by ChatGPT Application and PubMed digital library. We used "Chronic Obstructive Pulmonary Disease reports and evaluations" as input on ChatGPT application and PubMed digital library. Results are listed in Table 1.

As listed out in Table 1, six test and evaluation methods were reported by ChatGPT application. We retrieved information about these tests and evaluation method from six articles. The articles were selected randomly from PubMed digital library. None of the articles report all tests and

Table 1 COPD tests and evaluation methods

ChatGPT	Spirometry	Imaging tests	Arterial blood gas Test	Six-minute walk test	Quality of life questionnaires	Chest auscultation
Ammous et al. [10]	No	No	No	Yes	Yes	No
Cavaillès et al. [11]	Yes	No	No	Yes	Yes	No
Bos et al. [12]	Yes	No	No	No	Yes	No
Bestall et al. [13]	No	No	No	No	Yes	No
Alshamari et al. [14]	Yes	No	Yes	No	No	No
Moran-Mendoza et al. [15]	Yes	No	No	No	Yes	Yes
Vogelmeier et al. [16]	Yes	No	Yes	No	Yes	No

evaluation methods together. Studies [10, 12, 14] reported only two tests and evaluation methods, while articles [11, 15, 16] highlighted three kinds of test and evaluation methods. Study [13] reports only one test and evaluation method of COPD. Compared to all these studies, ChatGPT reportedly shows all these tests and evaluation methods together. This indicates that ChatGPT can effectively help physicians to choose the appropriate method among the reported test and evaluation methods by ChatGPT.

Conclusions

The widespread use of Open AI application in healthcare is inevitable. COPD is widely spread among the individuals worldwide. Developing and under-developed countries are facing more challenges of this disease due to lack of support and resources. ChatGPT can effectively provide invaluable information and guide healthcare providers to diagnose and manage the COPD among people at all age levels. Overall, the performance of ChatGPT application is better compared to the response from digital libraries. This paper has some implications for researchers to extend the role of ChatGPT application to cope with COPD and other diseases by timely information.

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Declarations

Conflict of interest No benefits in any form have been or will be received from a commercial party related directly or indirectly to the subject of this manuscript.

References

1. Benzo, R. P., B. Abascal-Bolado, and M. M. Dulohery. Self-management and quality of life in chronic obstructive pulmonary disease (COPD): the mediating effects of positive affect. *Patient Educ. Couns.* 99(4):617–623, 2016.
2. Gott, M., et al. Barriers to advance care planning in chronic obstructive pulmonary disease. *Palliat. Med.* 23(7):642–648, 2009.
3. Prasad, B. Chronic obstructive pulmonary disease (COPD). *Int. J. Pharm. Res. Technol.* 10(1):67–71, 2020.
4. Lortet-Tieulent, J., et al. International trends in COPD mortality, 1995–2017. *Eur. Respir. J.* 54(6):1901791, 2019.
5. Halpin, D. M., et al. It is time for the world to take COPD seriously: a statement from the GOLD board of directors. *Eur. Respir. Soc.* 54:1900914, 2019.
6. Cascella, M., et al. Evaluating the feasibility of ChatGPT in healthcare: an analysis of multiple clinical and research scenarios. *J. Med. Syst.* 47(1):1–5, 2023.
7. Biswas, S. S. Role of chat gpt in public health. *Ann. Biomed. Eng.* 2023:1–2, 2023.
8. Hasnain, M. ChatGPT applications and challenges in controlling monkey pox in Pakistan. *Ann. Biomed. Eng.* 2023. <https://doi.org/10.1007/s10439-023-03231-z>.
9. Mathews, A. M., et al. Hypercapnia in advanced chronic obstructive pulmonary disease: a secondary analysis of the National Emphysema Treatment Trial. *Chronic Obstr. Pulm. Dis.* 7(4):336, 2020.
10. Ammous, O., et al. Inspiratory muscle training, with or without concomitant pulmonary rehabilitation, for chronic obstructive pulmonary disease (COPD). *Cochrane Database Syst. Rev.* 2023. <https://doi.org/10.1002/14651858.CD013778.pub2>.
11. Cavaillès, A., et al. Comorbidities of COPD. *Eur. Respir. Rev.* 22(130):454–475, 2013.
12. Bos, L. D., P. J. Sterk, and S. J. Fowler. Breathomics in the setting of asthma and chronic obstructive pulmonary disease. *J. Allergy Clin. Immunol.* 138(4):970–976, 2016.
13. Bestall, J., et al. Usefulness of the Medical Research Council (MRC) dyspnoea scale as a measure of disability in patients with chronic obstructive pulmonary disease. *Thorax.* 54(7):581–586, 1999.
14. Alshamari, A. H. I., et al. Assessment of thyroid function tests in patients with chronic obstructive pulmonary disease. *J. Med. Life.* 15(12):1532, 2022.
15. Moran-Mendoza, O., T. Ritchie, and S. Aldhaheeri. Fine crackles on chest auscultation in the early diagnosis of idiopathic pulmonary fibrosis: a prospective cohort study. *BMJ Open Respir. Res.* 8(1):e000815, 2021.
16. Vogelmeier, C. F., et al. Goals of COPD treatment: focus on symptoms and exacerbations. *Respir. Med.* 166:105938, 2020.

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